

COMPLETE PROJECT PRESENTATION

United Lane System

A unified logistics operations platform: Fuel Service, Safety, Statistics, Driver, and Admin in one working system.

5

role-based workspaces

12+

main API areas

10

presentation pages

- Frontend: React + Vite, responsive workspaces, maps, and tables.
- Backend: FastAPI + SQLAlchemy + JWT, PostgreSQL/SQLite, Motive refresh.
- Integrations: Motive, TomTom, AI/OpenRouter, Telegram bot, Railway, Netlify.



Product Map

Each user enters the workspace that matches the role, while the backend verifies API permissions.

01 Admin

- Users, roles, bans
- Password resets and activity overview
- Last-admin protection

02 Fuel Service

- Loads, routing, tracking
- Fuel approvals and reconciliation
- Profitability, Full Road, Chat

03 Statistics

- Deep truck analytics
- Motive snapshots and archive trend
- MPG, mileage, faults, fuel

04 Safety

- Fleet safety, notes, documents
- Investigations and shift briefs
- AI chat, service/emergency tools

05 Driver

- Mobile-first workspace
- Linked to a Motive truck
- Fuel, route, service, chat

SHARED SYSTEM

Single Sign-In

- Department selection on auth page
- Office accounts are created by Admin
- Driver registration uses truck matching

Single Backend

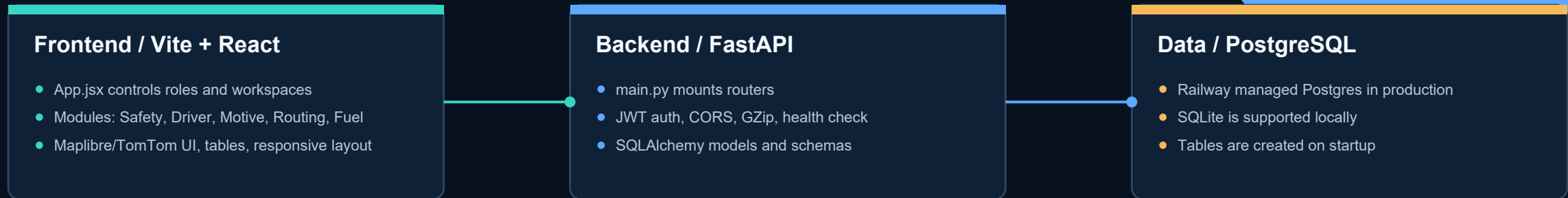
- All routes live under /api
- JWT access token
- department checks and ban checks

Single Data Layer

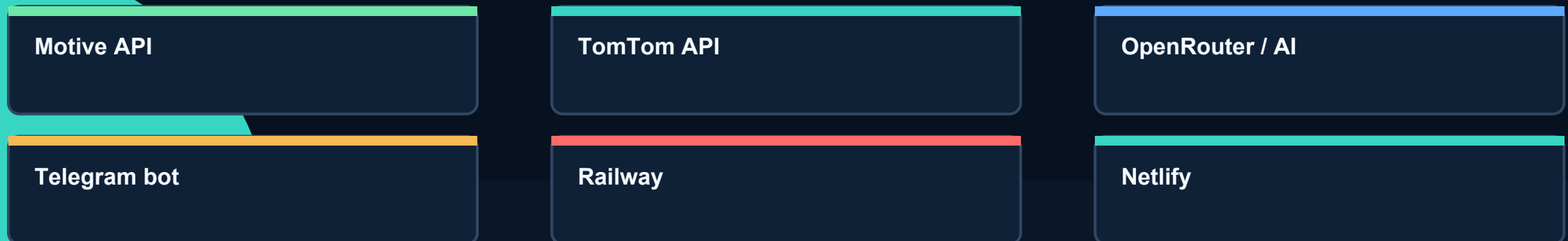
- Motive snapshots, loads, route history
- fuel authorizations, safety docs
- team chat and activity events

System Architecture

A monorepo with separate backend and frontend apps: local development, Railway API, Netlify UI, and PostgreSQL.



INTEGRATIONS AND RUNTIME



Fuel Service: Main Operations Center

Workspace for dispatch and fuel coordinators: from loads and maps to approvals and lane margin analysis.

01 Dashboard

- Load counts, active loads, low fuel, review items, miles left

02 Loads

- Dispatch table: driver, truck, status, MPG, fuel %, pickup, stops, delivery, pay inputs

03 Tracking

- Motive fleet data: status, driver, location, movement, stale data, low fuel, faults

04 Routing

- Route A -> B, live truck fuel/location, branded fuel stops

05 Approvals

- gallons, amount, price cap, expiration, sent/used/violated, reconciliation

06 Team + Settings

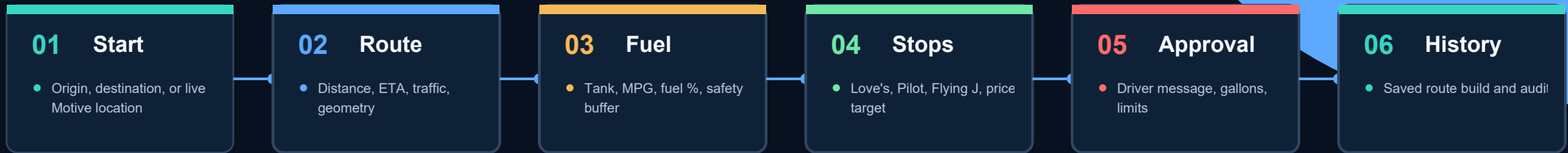
- Team Chat, workspace theme, font/background controls

FUEL WORKSPACE TABS

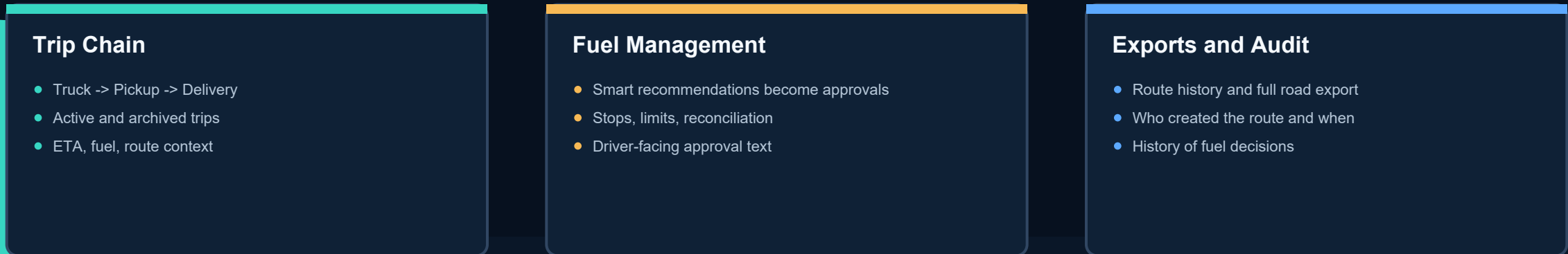
[Dashboard](#)[Tracking](#)[Statistics](#)[Profitability](#)[Full Road](#)[Routing](#)[Route History](#)[Approvals](#)[Loads](#)[Team Chat](#)[Settings](#)

Routing Works Like an Operations Flow

The system connects truck, origin, destination, fuel status, station data, and approval workflow.



FULL ROAD WORKSPACE



Statistics: Dates, Telematics, and Trends

Analytics are built on Motive snapshots, saved history, and clean data-date display.

Fleet

Total / located / online vehicles

Status

Moving, stopped, stale states

Truck

MPG, fuel %, odometer, mileage

Risk

Faults, incidents, driver context

Trends

Daily stats and archive growth

Analytics View

- Fleet dashboard with total, located, moving, stopped, online, stale, active drivers
- Truck detail: location, telemetry, driver, fuel, speed, odometer
- Filters by moving, stopped, stale, low fuel, faults, truck/driver/location

Data Freshness

- Dates and timestamps stay near data so users know freshness
- Motive background refresh lowers wait time
- Exports and snapshots help after-shift review

Safety: Control, Documents, and Support

Safety workspace brings compliance, fleet safety, service/emergency tools, and AI support together.

01 Safety Notes

- Private notes for safety users
- Saved on backend

02 Documents

- Upload, category, summary
- Issues, actions, excerpt, AI analysis

03 Investigations

- Facts, evidence, questions
- Owner, due date, severity, outcome

04 Shift Briefs

- Handoff notes
- Checklist, action items, shift status

05 Service Tools

- Service centers and emergency options
- Driver/Safety support map

06 Safety AI Chat

- Operational safety questions
- Image attachments where supported

EXPORTS / HANDOFF / SERVICE

Investigations export, shift brief export, risky people export, Motive-based safety fleet snapshot, and emergency/service tools for Safety and Driver.

People and Daily Work

The project covers access control, driver mobile flow, and shared communication.

Driver Workspace

- Mobile-first interface
- Truck number + Motive vehicle match
- Route, fuel, service, emergency support
- No extra Admin/Fuel access

Admin Panel

- Create/edit Admin, Fuel, Statistics, Safety, Driver accounts
- Ban/unban, reset passwords, protected deletes
- Overview: users, routes, fuel auth, team messages, safety docs

Team Chat

- Shared rooms for departments
- Reply, edit own messages, delete own messages
- Safety/Admin moderation and search

AUTH PAGE AND UX

Appearance Controls

- Auth page has font/background settings

Loading UX

- Centered larger loaders show when data is loading

The Backend Holds Rules and History

Critical restrictions live in the backend: auth, roles, bans, persistence, refresh, and integrations.

Security Model

- JWT authentication for protected routes
- Password hashes, not plain passwords
- Department checks and banned-user blocks

Data Models

- Users, activity events, loads, planner items
- Motive snapshots, daily stats, api connections
- Routes, full road trips, fuel authorizations, safety docs, chat

Runtime Config

- DATABASE_URL, SECRET_KEY, CORS_ORIGINS
- MOTIVE/TOMTOM/API/AI env variables
- Snapshot TTL, background refresh, gzip, workers

DEPLOY PATH

1. Railway DB

- Managed PostgreSQL
- DATABASE_URL reference

2. Railway API

- backend/ root
- FastAPI + worker/bot

3. Netlify UI

- frontend build
- VITE_API_URL=/api

4. Health

- /api/health
- docs at /docs

What the Project Already Contains

United Lane System already works as a full private operations platform.

Operational Value

- Less switching between sheets, maps, Motive, and chats
- Fuel decisions connect to route, limits, and reconciliation
- Safety and Driver get tools without extra permissions

Technical Base

- React/Vite frontend, FastAPI backend, SQLAlchemy persistence
- PostgreSQL production path, SQLite local path
- Role-based API protection, cache workers, exports, integrations

Ready to Grow

- Railway/Netlify deployment is documented
- Motive snapshots and route history provide audit base
- Telegram bot extends driver workflow outside browser

NEXT IMPROVEMENTS

- Lock production secrets and CORS for real domains before release.
- Validate real Motive/TomTom credentials and background refresh under load.
- Add smoke tests for auth, loads, route-assistant, motive fleet, fuel-authorizations, safety.
- Keep data dates next to metrics in all statistics and integration blocks.

PDF READY

10 pages about the full project